Assignment for Week 1

Author: Your name here!

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Assignment Part I

3.13 Let A, B, C be subsets of a set X. Which of the expressions $(A \cap B) \cup (A \cap B^c)$, $(A \cap B) \cup (A \cap B^c)^c$, $(A \cap B) \cup (A^c \cup C)^c$ can be simplified, and how? $(A \cap B) \cup (A^c \cup C)^c$

3.4.2 Let g be the mapping from $\mathbb{R}^2$ to $\mathbb{R}^2$ defined by

$$g(x, y) = (y, -x),$$

and let the mapping h be as in Exercise 3.4.1. Find the image of (x, y) under each of the composite mappings $g \circ h$ and $h \circ g$.

31.1.1 As in the text, let $\mathbb{Q}$ be the set of all rational numbers. If $x \in \mathbb{Q}$, which of the following statements (P, Q, R, S) imply which of the others?

P: $x \geq 0$

Q: $\exists w \in \mathbb{Q}$ such that $x = w^2$

R: $nx > -1 \ \forall n \in \mathbb{N}$

S: $\exists n \in \mathbb{N}$ such that $nx > -1$.

5.1.3 In each of the following cases, state whether the sequence $\{u_n\}$ tends to a limit, and find the limit if it exists:

(a) $u_n = 1 + \frac{1}{2}n$

(b) $u_n = 1 - \frac{1}{2}n$

(c) $u_n = (\frac{1}{2})^n$

(d) $u_n = -(\frac{1}{2})^n$

5.2.1 (p. 87) Find the sum of the natural numbers from 1 to 100 inclusive.

5.4.2 (p. 98) Let

Is this function continuous (a) at $x = 0$, (b) at $x = 1$? Is it a continuous function?

Assignment Part II

Package Loading

```
library(dplyr, warn.conflicts = FALSE)
library(ggplot2)
library(nycflights13)
```

(1) What are the types of data that serve as the argument and value of the following functions?

- `nchar()`
- `log()`
- `paste()`
- `is.na()`
- `str()`

(2) What was the longest physical distance taken out of New York in 2013?

(3) In R, we can use scientific notation. 2e3 is 2000. Why is log(1e10)-log(1e7) the same as log(1e3)?

(4) Without running these commands, what are the values of C, D, and E?

```
A <- "Iowa"
B <- "Kansas"

C <- !((A == "Kentucky") | (B == "Kansas"))

D <- ((nchar(A) > 2) & is.na(B)) | (!is.na(A))

E <- ifelse( C, D, A)
```

(5) Fix (with minimal changes) the following code snippets to produce TRUE. Hint, use ?, and str().

```
# 3=<3

# length(22[c(1,1,1,1)][-1])==4

# mean[rnorm(100)]

# sum(ifelse(runif(10)>.9, "1","0"))>5


# df<-data.frame(GreekLetters=c("Alpha", "Beta", "Gamma"))
# nchar(df$GreekLetters)[1]>3
```

(6) Fix the following ggplot commands:

```
votes2016<-c(-5, -4, -3, -10, 0, 3, 4, 0, 5)
votes2018<-c(0, -2, -8, -8, -9, 12, 13, 5, 3)
Primary<-c("T","C","T","T","T","C","C","C","C")
votes2020<-c(-4, -2, -5, -8, -1, 2, 8, 5, 3)

df<-data.frame(votes2016=votes2016, votes2018=votes2018, primary=Primary, votes2020=votes2020)

ggplot(df, aes(x = "votes2016", y = "votes2018")) +
  geom_point()

ggplot(df, aes(x = votes2016, y = votes2018)) +
  geom_density()

ggplot(df, aes(x = votes2016, y = votes2018)) +
  geom_point(color=primary)

ggplot(df, aes(x = votes2016, y = votes2018)) +
  geom_point(aes(color=primary))+geom_point(x=4, y=4,aes(color="blue"),shape= 3)

ggplot(df) +geom_point(aes(x = votes2016, y = votes2018))+geom_smooth()

ggplot(df) +geom_point(aes(x = votes2016, y = votes2018))+
  geom_col(filter(df, primary=="T"), aes(x = votes2016, y = votes2020) )
```

(7) Plot an interesting subset of the data Flights data. Use the Rmarkdown fig.height and fig.width to make it look nice.

(8) Write your own "quantile" function which gets the median and the 95th percentile of a vector (without using the base r quantile). For example, here is a function which tells you what proportion of inputs are five digit numbers

```
zipcheck<-function(zip){
  zipn<-as.numeric(zip)
  isfivedigit <- abs(zipn)>9999 &  abs(zipn) < 100000
  propout<-sum(isfivedigit, na.rm=T)/ length(isfivedigit)
  return(propout)
  }

zipcheck(c(10,10001,60605,-40304))
```

(9) What does `df` look like when it is printed out?

```
df <- data.frame(dogheight=c("10", "20", "30"), dogwidth=c("5", "3", "1"), stringsAsFactors = FALSE)

df[df$dogwidth==3,1] <- NA
```

(10) *True* or *False*

(a) A small p-value, such as .001 or .0001, means that our estimate is unlikely to have arisen from the null hypothesis.
(b) The small errors distribution is a reasonable approximation for the distances traveled by flights out of New York City.
(c) $f(x) = 2x$ for $0 \leq x \leq 1$ is a legitimate probability density function.
(d) The probability that a single draw from a standard normal distribution is 0 is 0.